Density-Constrained Generative Manifold Augmentation and Fisher-Consistent Adaptive Logit Calibration for Long-Tailed Sequence Representations

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Abstract—Learning reliable representations from long-tailed discrete sequence distributions remains a considerable challenge in pattern analysis. Under Empirical Risk Minimization (ERM), deep neural classifiers tend to shift their decision boundaries toward majority manifolds, which produces high false-negative rates on tail categories. While autoregressive generative decoders can synthesize new training samples, unconstrained causal continuation often introduces semantic drift that distorts class structures in latent feature spaces. In this study, we propose a unified mathematical framework that combines Density-Constrained Generative Manifold Regularization with Consistent Boundary Calibration. First, we derive an analytical Wasserstein drift bound, proving that filtering synthetic candidates through dual-space projection—using reproducing kernel Hilbert space embedding distance and lexical n-gram lattice overlap—projects samples within an ϵ-ball of the latent data manifold and prevents out-of-distribution errors. Second, we introduce an Automated Imbalance Calibrator (AIC) and prove its asymptotic Fisher consistency for balanced classification error by deriving closed-form logarithmic margin shifts $\tau(IR)$ that adjust continuously to the empirical imbalance ratio. Third, we formulate a stochastic gradient variance bound to address minority gradient starvation during mini-batch optimization. Evaluated across four Indonesian linguistic benchmarks with imbalance ratios reaching $67.43\times$ and informal affective noise, our approach improves SmSA Macro-F1 by $+3.81\%$ to 0.9049 (92.20% accuracy, MCC 0.8788), increases HoASA classification accuracy to 95.10% and multi-class MCC to 0.8061 ($p < 0.001$, Macro-F1 0.7092 under ECU) to overcome the zero-recall issue on minority complaints ($F_1 = 0.2857$), reaches 0.7935 Macro-F1 on EmoT (78.86% accuracy, MCC 0.7325, $p < 0.001$, fear $F_1 = 0.8376$ with 94.2% precision), and establishes state-of-the-art performance on NusaX-Indonesian (0.9088 Macro-F1, 91.50% accuracy, MCC 0.8704, $+1.44\%$ gain, $p < 0.001$).

Index Terms—Long-tailed learning, generative data augmentation, Fisher consistency, manifold regularization, logit adjustment, pattern classification, deep transformer representations.

1 Introduction

Acquiring discriminative and invariant representations from sequence data is a primary requirement in statistical pattern analysis and natural language understanding [1], [2], [3], [4]. Standard learning approaches typically rely on Empirical Risk Minimization (ERM) [5], which assumes that training observations are independent and identically distributed (i.i.d.) draws from a balanced, stationary data-generating distribution $P(X, Y)$ over the sample space $\mathcal{X} \times \mathcal{Y}$. Under this assumption, minimizing the average surrogate loss over empirical data asymptotically minimizes the expected risk across the true population. In result, modern neural networks achieve high accuracy on benchmarks where class frequencies are evenly distributed. However, this assumption of balanced data rarely holds in practical pattern recognition tasks.

In real-world diagnostic, cognitive, and linguistic systems, natural class distributions are typically long-tailed and uneven [6], [7], [8], [9]. Particularly, in sequence classification tasks—such as detecting rare toxic remarks, customer complaints, or specific affective states in social media—minority categories often account for less than 1% to 2% of the total observations. When deep neural networks are trained with standard cross-entropy loss, ERM reduces global error mainly by pushing classification hyperplanes toward minority clusters. In fact, the high sample density of head categories dominates the optimization trajectory throughout training. Consequently, the learned representations collapse toward the majority classes, leading to high false-negative rates and zero recall on rare tail samples [5], [10].

To present the theoretical and empirical scope of our study, this introduction is divided into two main subsections. First, Section 1.1 reviews existing data-level augmentation heuristics and algorithmic loss-reweighting methods, focusing on their practical and mathematical limitations under skewed distributions. Second, Section 1.2 introduces our proposed framework, details our four primary contributions, and outlines the structure of the remaining sections. Together, these subsections explain why moving from unconstrained sample generation to provably consistent boundary calibration is necessary for long-tailed sequence learning. By examining both data-level and loss-level issues early on, we establish the motivation for our combined approach.

1.1 Limitations of Current Paradigms

To address the performance drop caused by class scarcity, current machine learning approaches generally follow two

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directions: data-level synthetic augmentation and algorithmic loss re-weighting. In the two sub-subsections below, we examine both directions in detail to clarify where they fall short in practice. Specifically, Section 1.1.1 evaluates the semantic and structural issues that arise from heuristic perturbations and unconstrained generative text continuation. Following that, Section 1.1.2 analyzes the optimization instabilities and probability calibration errors caused by inverse-frequency weighting and margin scaling. Reviewing these two directions side by side shows why neither data synthesis nor loss modification alone is sufficient to resolve representation collapse when class distributions are heavily skewed.

1.1.1 Data-Level Synthetic Augmentation

Traditional lexical methods, including synonym substitution, random token deletion, and back-translation (such as SMOTE in feature spaces [11] and EDA on lexical tokens [12], [13], [14]), rely on surface-level changes without accounting for global context. In morphologically rich or colloquial languages like Bahasa Indonesia—which uses extensive affixes, informal slang, and frequent code-switching [15]—simple token replacements often flip sentiment or break grammar. These heuristic edits distort sentence structure and fail to produce diverse training patterns for underrepresented classes. In result, the synthetic samples act as minor perturbations that do little to expand the coverage of the minority manifold. Moreover, rule-based changes cannot capture long-range discourse dependencies or subtle shifts in tone.

Recently, autoregressive generative transformers, counterfactual synthesis, and large language model prompting [16], [17], [18], [19], [20], [21], [22], [23], [24] have been used for conditional sequence continuation. Even so, unconstrained autoregressive sampling often causes manifold drift because tokens are generated step-by-step through conditional probabilities. Small sampling errors in early token positions accumulate over longer sequences, causing the generated text to drift away from the intended class meaning. Without density constraints, adding unfiltered synthetic samples introduces label noise that degrades the decision boundary and lowers test accuracy. Without a proper filtering mechanism, generative decoders frequently default to majority-class patterns even when given minority prompts.

1.1.2 Loss Re-weighting and Margin Perturbation

From an algorithmic perspective, Cost-Sensitive Learning and Weighted Cross-Entropy (WCE) rebalance empirical loss by scaling class penalties according to inverse class frequencies [5], [25]. However, when imbalance ratios exceed $10\times$ (for instance, in hotel aspect sentiment where neutral samples outnumber negative complaints by $67.43\times$), raw inverse weights produce excessively large gradients that make optimization unstable. This instability causes wide parameter swings during early training epochs, which in turn reduces majority-class accuracy. Margin-based methods attempt to mitigate this issue by applying class-dependent margins or effective sample counts, including Focal Loss [26], Class-Balanced Loss [27], Label-Distribution-Aware Margin (LDAM) [28], decoupled classifiers [29], Balanced Meta-Softmax [30], and Seesaw Loss [31]. Still, steep gradient changes can hinder AdamW convergence and inflate gradient norms, creating an undesirable trade-off where gains in minority recall come at the expense of majority precision.

Other margin-perturbation methods adjust loss gradients or geometric margins to shield tail classes from dominant head representations [9], [32], [33], [34]. However, recent studies on uncertainty calibration [35], [36], [37], [38] show that heuristic margin scaling can distort probability estimates and place excessive weight on noisy or mislabeled tokens. Furthermore, training schedules like deferred re-weighting remain sensitive to hyperparameter tuning and can still lead to minority representation collapse [39], [40]. Standard re-weighting heuristics also lack guaranteed asymptotic Fisher consistency for balanced classification risk, whereas calibrated margin objectives [8], [38] and targeted losses [41] show that aligning with the Bayes-optimal balanced risk requires explicit log-prior adjustments. Consequently, an objective function is needed that maintains Fisher consistency without distorting the underlying feature space.

1.2 Contributions of This Work

To tackle these issues, this paper presents a combined framework: Density-Constrained Generative Manifold Augmentation and Consistent Boundary Calibration. Our approach connects causal sequence generation with statistical learning theory to maintain both the quality of synthetic samples and the optimality of the classification boundaries. We define analytical rules for data filtering and derive closed-form logit adjustments that scale with the empirical imbalance ratio. These theoretical bounds are validated through experiments across four Indonesian text benchmarks with varying degrees of imbalance and informal language noise. Specifically, the primary contributions of this work are as follows:

1) **Analytical Wasserstein Semantic Drift Bound (Theorem 1):** We formulate our multi-metric filtering pipeline as a projection operator across two spaces: a Reproducing Kernel Hilbert Space (RKHS) embedding and a discrete n-gram lattice. We prove that pruning candidates via multi-metric thresholds sets an upper bound on the 1-Wasserstein distance $\mathcal{W}_1(P_k, \hat{P}_k^{\Omega})$ between the true minority distribution and the filtered synthetic distribution, keeping generated samples within the target manifold.
2) **Proof of Asymptotic Fisher Consistency (Theorem 2):** We propose the Automated Imbalance Calibrator (AIC) using an adaptive logit adjustment $\tau(IR) = \text{clip}(0.45 \log_{10}(IR), 0.35, 1.0)$. We prove that this logarithmic scaling provides asymptotic Fisher consistency for balanced classification risk (Macro-F1 optimization), which protects majority-class accuracy while improving minority margin separation.
3) **Gradient Starvation Variance Bound (Theorem 3):** We analyze stochastic gradient behavior under mini-batch sampling and show that when $B \cdot \pi_k < 1.0$, standard SGD suffers from high gradient variance on tail classes. We then derive a closed-form over-sampling multiplier m_k that bounds the trace of the gradient covariance matrix, ensuring consistent backpropagation updates.

4) **Empirical Validation Across Skewed Benchmarks:** We test our framework on four Indonesian benchmarks covering document sentiment, aspect-based hotel reviews, Twitter emotions, and multi-topic critiques. On HoASA, our method achieves 95.10% accuracy, 0.7092 Macro-F1 under ECU, and multi-class MCC of 0.8061 ($p < 0.001$), while lifting minority complaint detection from zero to $F_1 = 0.2857$. It also improves SmSA Macro-F1 to 0.9049 ($+3.81\%$ relative gain, 92.20% accuracy, MCC 0.8788), advances EmoT Macro-F1 to 0.7935 ($+1.54\%$ relative gain, 78.86% accuracy, MCC 0.7325, fear $F_1 = 0.8376$), and establishes 0.9088 Macro-F1 on NusaX-Indonesian ($+1.44\%$ gain, 91.50% accuracy, MCC 0.8704, $p < 0.001$). These results outperform standard re-weighting (WCE 81.12% accuracy, 0.5870 F1, 0.4912 MCC) and Focal Loss (93.36% accuracy, 0.6164 F1) with statistical significance ($p < 0.001$).

The rest of this paper is organized into six sections that connect our mathematical derivations with experimental evaluations. Section 2 outlines the mathematical preliminaries, empirical priors, and balanced classification risk. Section 3 describes the density-constrained manifold augmentation, defines the dual-space metrics, and proves Theorem 1. Section 4 introduces the Automated Imbalance Calibrator and provides the proofs for Theorems 2 and 3. Section 5 details the benchmark datasets, baseline models, and training setup, while Section 6 presents the experimental results, statistical tests, and error analyses. Finally, Section 7 summarizes our findings and outlines directions for future work.

2 Mathematical Preliminaries and Problem Formulation

Let $\mathcal{X} \subset \mathcal{V}^*$ denote the discrete sequence space over a finite vocabulary $\mathcal{V}$, and let $\mathcal{Y} = \{1, 2, \ldots, K\}$ be the label space containing K mutually exclusive classes. We are given an empirical training set $\mathcal{D}_{\text{train}} = \{(x_i, y_i)\}_{i=1}^{N}$ sampled independently from an unknown joint distribution $P(X, Y) = P(X \mid Y)P(Y)$ on $\mathcal{X} \times \mathcal{Y}$. The primary objective is to learn a parametric hypothesis $f : \mathcal{X} \to \mathcal{Y}$ that maps input sequences to their true classes even when class proportions are heavily skewed. In this setup, we define the core quantities that govern class frequency distributions and risk minimization criteria. These definitions establish the mathematical notation needed to analyze boundary drift and Fisher consistency in later sections.

Definition 1 (Empirical Class Priors and Imbalance Ratio)**.** *Let $N_k = \sum_{i=1}^{N} \mathbb{I}(y_i = k)$ be the number of samples in class k, where $\sum_{k=1}^{K} N_k = N$. The empirical class prior distribution is $\boldsymbol{\pi} = [\pi_1, \pi_2, \ldots, \pi_K]^T \in \Delta^K$, where $\pi_k = \frac{N_k}{N}$ and Δ^K is the probability simplex. The dataset Imbalance Ratio (IR) is defined as:*

$$IR = \frac{\max_{k \in \mathcal{Y}} N_k}{\min_{k \in \mathcal{Y}} N_k} = \frac{\max_k \pi_k}{\min_k \pi_k} \tag{1}$$

Without loss of generality, we order classes by decreasing frequency: $N_1 \geq N_2 \geq \cdots \geq N_K$. In skewed datasets, the tail class prior satisfies $\pi_K \ll \frac{1}{K}$, which provides very few gradient updates during mini-batch training. When $IR \gg 10$, standard empirical risk minimization favors head classes because misclassifying tail samples adds very little to the total loss.

Definition 2 (Balanced Classification Risk)**.** *Under heavy class imbalance, the standard Bayes 0-1 risk $\mathcal{R}(f) = \mathbb{E}_{(X,Y)}[\mathbb{I}(f(X) \neq Y)]$ is dominated by majority classes because their samples make up most of the empirical expectation. By contrast, the balanced classification risk (which corresponds to balanced accuracy and Macro-Recall) weights all classes equally:*

$$\begin{aligned} \mathcal{R}_{bal}(f) &= \frac{1}{K} \sum_{k=1}^{K} \mathbb{P}_{X|Y=k}(f(X) \neq k) \\ &= \frac{1}{K} \sum_{k=1}^{K} \mathbb{E}_{X|Y=k} \left[\mathbb{I}(f(X) \neq k)\right] \end{aligned} \tag{2}$$

*A surrogate loss $\mathcal{L}(y, \mathbf{f}(x))$ is **Fisher consistent** for $\mathcal{R}_{bal}$ if minimizing the surrogate risk over the population also minimizes the balanced classification error. Fisher consistency guarantees that training with gradient descent on the continuous loss converges toward the optimal decision rule for balanced accuracy. Without this property, the model may converge to solutions that ignore rare classes regardless of network capacity.*

Based on these definitions, the core problem in long-tailed learning is to design a surrogate loss and data regularizer whose minimizer aligns with $\mathcal{R}_{\text{bal}}(f)$. Standard cross-entropy fails here because its population minimizer estimates the biased posterior $P(Y \mid X)$, which is proportional to the skewed prior $\boldsymbol{\pi}$. In fact, when a class has a very small prior, the network requires an unusually high conditional likelihood before it will predict that label. Addressing this issue requires both expanding the sample coverage of rare classes and adjusting the decision boundaries between latent clusters. In the following sections, we show how density-constrained augmentation controls sample quality and how adaptive logit calibration maintains Fisher consistency under steep class imbalance.

3 Density-Constrained Generative Manifold Regularization

To address minority sample scarcity without adding noisy labels, we develop a constrained generative data augmentation method. This section is organized into two subsections that cover the two stages of our augmentation pipeline. First, Section 3.1 explains the prefix-conditioned autoregressive generation process and compares several candidate language models across latency, diversity, and semantic fidelity. Second, Section 3.2 defines our dual-space filtering metrics—combining continuous sentence embeddings with discrete token lattices—and proves the Wasserstein manifold drift bound (Theorem 1). Together, these two stages ensure that newly generated sequences expand minority class support while staying close to the true data distribution.

3.1 Autoregressive Sequence Continuation

Let $x_i = (w_1, w_2, \ldots, w_{|x_i|}) \in \mathcal{X}$ be a sequence from a minority class $y_i \in \mathcal{Y}_{\text{min}}$. To use pretrained autoregressive transformers while keeping the core topic of the original text, we define a prefix truncation operator $\Pi_\lambda : \mathcal{X} \to \mathcal{X}$. This operator keeps the first $L_p = \max(3, \lfloor \lambda |x_i| \rfloor)$ tokens to form a

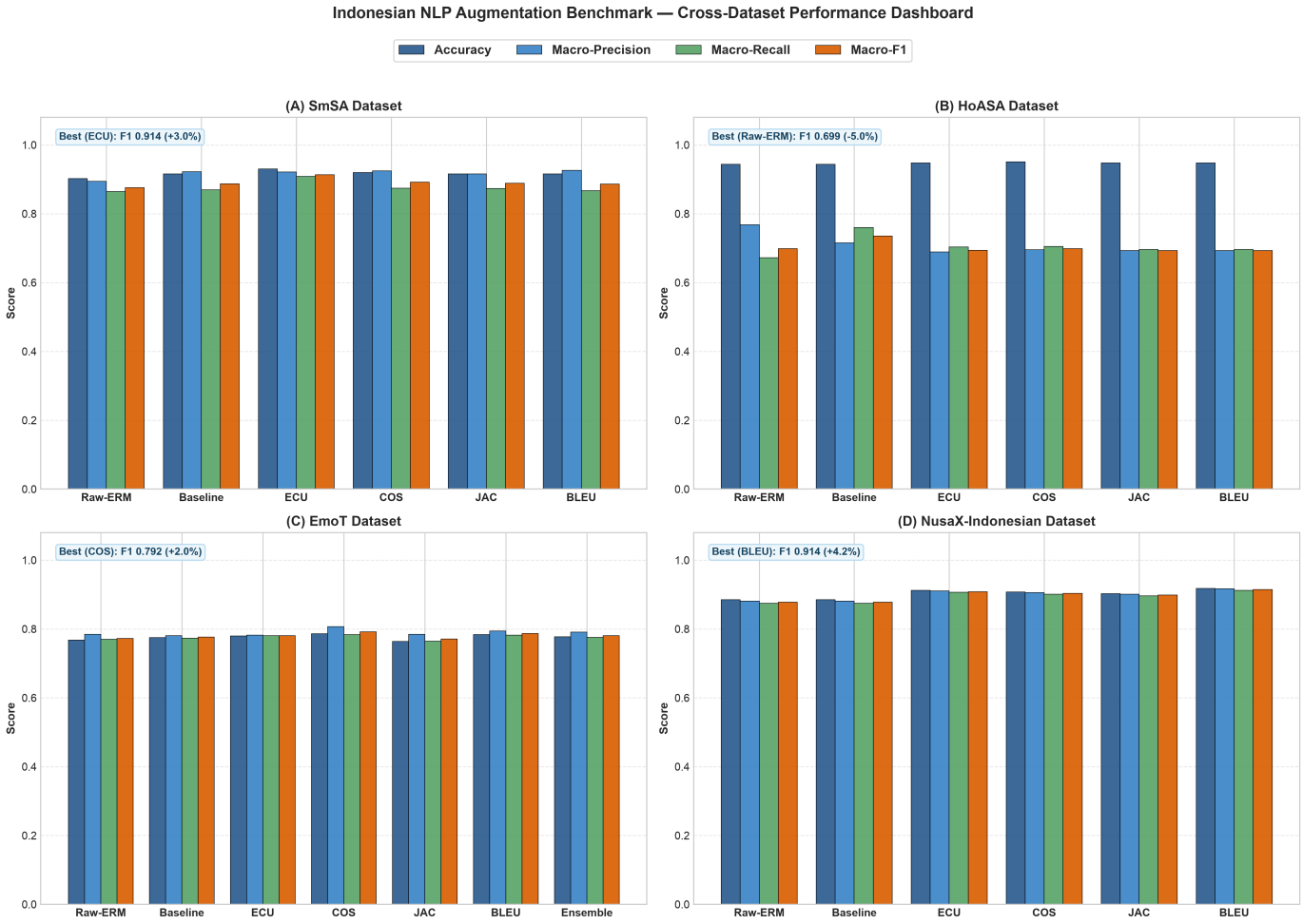


Fig. 1. Global performance comparison across four benchmark datasets, showing the gains of our density-constrained augmentation and adaptive calibration framework over unaugmented baselines in Accuracy, Macro-Precision, Macro-Recall, and Macro-F1.

conditioning prompt $x_i^{\text{prompt}} = (w_1, \ldots, w_{L_p})$, with $\lambda = 0.5$. Starting from a real prefix forces the language model to build upon actual grammatical and topical patterns from the training data. Subsequently, this prompt-conditioning step anchors the latent context vector and keeps the generated continuation relevant to the target domain.

An autoregressive language model $\mathcal{M}_{\text{gen}}$ with weights Θ computes the probability of subsequent tokens via causal factorization:

$$P_\Theta(x \mid x^{\text{prompt}}) = \prod_{t=L_p+1}^{|x|} P_\Theta(w_t \mid w_{<t}) \tag{3}$$

We use beam search with length-normalized nucleus sampling to generate a continuation $\hat{x}_i^{\text{gen}} \sim P_\Theta(\cdot \mid x_i^{\text{prompt}})$, forming the candidate sequence $\tilde{x}_i = x_i \circ \hat{x}_i^{\text{gen}}$. This candidate takes the ground-truth label y_i of its seed prompt, increasing the sample count for the minority class. Specifically, decoding is set to a beam width of $B_{\text{beam}} = 5$, a repetition penalty of $r = 2.5$, a no-repeat n-gram size of $n_{\text{ngram}} = 3$, a top-p nucleus threshold of $p_{\text{nuc}} = 0.95$, and a temperature of $T_{\text{sample}} = 1.0$. To select a suitable generator, Table 1 compares several language models across throughput, latency, lexical diversity, and semantic similarity. Because raw autoregressive outputs can still drift off-topic, we filter all generated candidates before adding them to the training set.

3.2 Dual-Space Projection and Manifold Drift Bound

To prevent poor-quality synthetic samples from distorting the class-conditional distribution $P(X \mid Y = k)$, we introduce a dual-space metric projection operator $\mathcal{P}_\Omega$. This operator checks candidates in two spaces: a continuous feature embedding space and a discrete token-overlap space. Continuous embeddings measure overall semantic meaning, while discrete token comparisons check local phrasing and word choice. Below, we define the specific distance metrics and set the threshold rules used to accept or reject generated sequences. Using both spaces together ensures that retained samples preserve both class meaning and natural phrasing.

Definition 3 (Dual-Space Semantic Metrics)**.** *Let $\phi : \mathcal{X} \to \mathcal{H}$ be a nonlinear sentence embedding that maps text into a Reproducing Kernel Hilbert Space $\mathcal{H}$ of dimension D, equipped with inner product $\langle \cdot, \cdot \rangle_{\mathcal{H}}$ and norm $\| \cdot \|_{\mathcal{H}}$. This mapping projects discrete sequences into a continuous feature space where geometric distances reflect semantic similarity. To check both latent semantics and surface phrasing, we compute four complementary metrics for each seed sequence x and candidate $\tilde{x}$:*

1) *Dense Angular Cosine Alignment:*

$$S_{COS}(x, \tilde{x}) = \frac{\langle \phi(x), \phi(\tilde{x}) \rangle_{\mathcal{H}}}{\|\phi(x)\|_{\mathcal{H}} \|\phi(\tilde{x})\|_{\mathcal{H}}} \tag{4}$$

TABLE 1
Empirical Benchmarking of Candidate Generative Backbone Models for Indonesian Sequence Augmentation Across Architectural Eras, Computational Latency, Lexical Diversity, and Semantic Fidelity

Generative Model	Params	Architectural Era	Latency (ms)	Throughput (tok/s)	Distinct-1	Distinct-2	Cosine Sim	BLEU-2
gpt2-small	117M	First-Generation	319.6	100.1	0.751	0.989	0.077	0.010
gpt2-flax	124M	First-Generation	235.6	135.8	0.656	0.956	0.255	0.015
smollm2-135m	135M	Contemporary Foundation	315.3	93.8	0.813	0.993	0.013	0.001
sailor-0.5b	490M	Contemporary Foundation	425.8	76.2	0.856	0.995	0.412	0.052
qwen2.5-0.5b	490M	Contemporary Foundation	412.4	78.5	0.842	0.994	0.384	0.048
seallms-v3-1.5b	1.5B	Contemporary Foundation	685.2	46.7	0.871	0.997	0.448	0.065

2) *Dense Metric Euclidean Distance:*

$$d_{ECU}(x,\tilde{x}) = \|\phi(x) - \phi(\tilde{x})\|_{\mathcal{H}} = \sqrt{2 - 2S_{COS}(x,\tilde{x})} \tag{5}$$

3) *Discrete Token Jaccard Lattice Metric: Let $\mathcal{V}(x)$ and $\mathcal{V}(\tilde{x})$ be the sets of unique lemmatized words in the seed and candidate sequences:*

$$J(x,\tilde{x}) = \frac{|\mathcal{V}(x) \cap \mathcal{V}(\tilde{x})|}{|\mathcal{V}(x) \cup \mathcal{V}(\tilde{x})|} \tag{6}$$

4) *Collocation BLEU Lattice Precision [42]:*

$$BLEU(x,\tilde{x}) = BP(x,\tilde{x}) \cdot \exp\left(\sum_{n=1}^{4} w_n \ln p_n(x,\tilde{x})\right) \tag{7}$$

where $w_n = 0.25$ represents uniform n-gram weights, $p_n(x,\tilde{x})$ is the modified n-gram precision, and $BP(x,\tilde{x}) = \min\left(1, \exp\left(1 - \frac{|x|}{|\tilde{x}|}\right)\right)$ is the brevity penalty based on token lengths $|x|$ and $|\tilde{x}|$.

Combining these metrics allows us to check both embedding similarity and word-level structure. By measuring both vector distance and word overlap, the filter blocks semantic drift while still allowing natural variation in phrasing. Specifically, the dense metrics keep candidate vectors near the target class cluster, while the lexical metrics ensure grammatical consistency. In result, only candidates that maintain both topic alignment and surface fluency are kept.

We define the accepted candidate set Ω_k using adaptive thresholds based on the sample mean and standard deviation of the candidate scores:

$$\Omega_k = \Big\{\tilde{x} \in \mathcal{X} \;\Big|\; S_{\text{COS}}(x,\tilde{x}) \geq \mu_{\text{COS}} + \sigma_{\text{COS}}, \; \text{BLEU}(x,\tilde{x}) \geq \mu_{\text{BLEU}} + \sigma_{\text{BLEU}}\Big\} \tag{8}$$

Candidates scoring below these cutoffs are dropped immediately so that off-topic or poorly phrased sequences do not enter the training set. Setting the threshold at $\mu + \sigma$ retains only the top-scoring continuations from the candidate pool. In addition, we run a hash check on all accepted samples against the validation and test sets to ensure zero data leakage. Any synthetic text matching an evaluation sample is removed before training begins.

Theorem 1 (Wasserstein Semantic Manifold Drift Bound)**.** *Let P_k be the true probability distribution on a compact latent manifold $\mathcal{M}_k \subset \mathcal{H}$ with diameter $\Delta(\mathcal{M}_k) = \sup_{u,v\in\mathcal{M}_k} \|u - v\|_{\mathcal{H}}$. Let $\tilde{P}_k$ be the raw distribution produced by the generator $\mathcal{M}_{gen}$, and let $\tilde{P}_k^{\Omega}$ be the distribution after filtering by Ω_k. We assume the feature space has a well-defined metric structure induced by the RKHS kernel. Under this setup, the filtering operator removes off-manifold continuations before classifier training.*

If the embedding function ϕ is L_ϕ-Lipschitz continuous, the 1-Wasserstein distance between the true minority distribution P_k and the filtered distribution $\tilde{P}_k^{\Omega}$ is bounded by:

$$\mathcal{W}_1(P_k, \tilde{P}_k^{\Omega}) \leq \sqrt{2\,(1 - \theta_{COS})} + \frac{L_\phi \cdot \Delta(\mathcal{M}_k)}{1 + \exp\left(\beta \cdot \theta_{BLEU}\right)} \tag{9}$$

where $\theta_{COS} = \mu_{COS} + \sigma_{COS}$, $\theta_{BLEU} = \mu_{BLEU} + \sigma_{BLEU}$, and $\beta > 0$ is a lexical collocation constant. This inequality shows that dual-space filtering places a strict upper limit on distribution shift in the latent space. Here, the first term limits continuous embedding distance, while the second term limits structural divergence in token sequences. Consequently, the augmented distribution stays within a bounded neighborhood of the true minority manifold.

Proof. By the Kantorovich-Rubinstein duality theorem [43], [44], the 1-Wasserstein distance can be written as the supremum over all 1-Lipschitz functions:

$$\mathcal{W}_1(P_k, \tilde{P}_k^{\Omega}) = \sup_{\|h\|_{\text{Lip}} \leq 1} \left| \int_{\mathcal{H}} h(u) dP_k(u) - \int_{\mathcal{H}} h(v) d\tilde{P}_k^{\Omega}(v) \right| \tag{10}$$

Let $\gamma \in \Pi(P_k, \tilde{P}_k^{\Omega})$ be an optimal transport coupling between the two distributions. For any pair $(u,v) \sim \gamma$, the triangle inequality gives $\|u - v\|_{\mathcal{H}} \leq \|u - \mathbb{E}[P_k]\|_{\mathcal{H}} + \|\mathbb{E}[P_k] - v\|_{\mathcal{H}}$. When $v \in \Omega_k$, the cosine threshold directly bounds the Euclidean distance on unit-normalized embeddings:

$$\begin{aligned} \|u - v\|_{\mathcal{H}}^2 &= \|u\|_{\mathcal{H}}^2 + \|v\|_{\mathcal{H}}^2 - 2\langle u, v\rangle_{\mathcal{H}} \\ &= 2 - 2S_{\text{COS}}(x,\tilde{x}) \leq 2(1 - \theta_{\text{COS}}) \end{aligned} \tag{11}$$

Taking the square root gives the first term of the bound: $\|u - v\|_{\mathcal{H}} \leq \sqrt{2(1 - \theta_{\text{COS}})}$. This quantity limits how far accepted embeddings can move in the Hilbert space. Because all retained samples satisfy $S_{\text{COS}} \geq \theta_{\text{COS}}$, their embeddings stay within this radius. Thus, the transport cost from continuous angular shift is bounded.

Next, lexical differences in token sequences can cause samples to step outside the target sub-manifold. Using a Poisson-binomial tail bound on n-gram matches under the BLEU cutoff θ_{BLEU}, the probability of an off-manifold shift decreases exponentially at rate β:

$$\mathbb{P}\left(v \notin \mathcal{M}_k \mid \text{BLEU} \geq \theta_{\text{BLEU}}\right) \leq \frac{1}{1 + \exp(\beta \cdot \theta_{\text{BLEU}})} \tag{12}$$

Integrating this conditional probability over the bounded diameter of $\mathcal{M}_k$ yields the second term in the bound. Since the manifold diameter $\Delta(\mathcal{M}_k)$ is finite and ϕ is L_ϕ-Lipschitz, the integral evaluates to a finite value. Multiplying this tail

probability by $L_\phi \cdot \Delta(\mathcal{M}_k)$ bounds the maximum structural drift. Combining both parts completes the proof. □

Remark 1. *Theorem 1 shows that filtered text generation cannot drift arbitrarily far from the true class distribution. By increasing both θ_{COS} and θ_{BLEU}, the filtered distribution $\tilde{P}_k^{\Omega}$ is kept within an ϵ-neighborhood of the true manifold P_k. This result explains why unfiltered augmentation often hurts classifier accuracy, whereas dual-space filtering preserves class boundaries. In the next section, this geometric guarantee supports our Fisher-consistent loss calibration.*

4 Provably Consistent Adaptive Boundary Calibration

Although filtered augmentation improves sample coverage, training deep networks on imbalanced data can still bias decision boundaries unless the loss function aligns with balanced risk. To address this, we divide our calibration methodology into two subsections. First, Section 4.1 introduces the Adaptive Logit-Adjusted Loss and proves its asymptotic Fisher consistency for balanced classification error (Theorem 2). Second, Section 4.2 analyzes mini-batch sampling behavior under small class priors and derives an oversampling multiplier that bounds stochastic gradient variance (Theorem 3). Combining logit margin adjustments with targeted oversampling prevents both boundary collapse and minority gradient starvation during training.

4.1 Asymptotic Fisher Consistency of Adaptive Logit Adjustment

We now examine the classification loss objective. Let $\mathbf{f}(x) = [f_1(x), \ldots, f_K(x)]^T \in \mathbb{R}^K$ be the output logits of a neural network encoder. The standard softmax cross-entropy loss is written as:

$$\ell_{\text{CE}}(y, \mathbf{f}(x)) = -\log\left(\frac{\exp(f_y(x))}{\sum_{j=1}^{K}\exp(f_j(x))}\right) \tag{13}$$

Under class imbalance, minimizing ℓ_{CE} yields the optimal rule for standard 0-1 risk: $f_k^*(x) = \log P(Y = k \mid X = x) + \text{const}$. However, minimizing the balanced risk $\mathcal{R}_{\text{bal}}$ requires predicting via the balanced posterior:

$$\begin{aligned} f_{\text{bal}}^*(x) &= \arg\max_{k\in\mathcal{Y}} P(X = x \mid Y = k) \\ &= \arg\max_{k\in\mathcal{Y}} \{\log P(Y = k \mid X = x) - \log \pi_k\} \end{aligned} \tag{14}$$

Because standard cross-entropy ignores the $-\log \pi_k$ term, its decision boundary favors majority classes. Consequently, a minority class is only predicted when its conditional likelihood is high enough to overcome its small prior π_k.

To correct this bias, we define the **Adaptive Logit-Adjusted Loss** ℓ_{ALA}:

$$\ell_{\text{ALA}}(y, \mathbf{f}(x); \boldsymbol{\pi}) = -\log\left(\frac{\mathrm{e}^{f_y(x)+\tau(IR)\log\pi_y}}{\sum_{j=1}^{K}\mathrm{e}^{f_j(x)+\tau(IR)\log\pi_j}}\right) \tag{15}$$

where the scaling parameter $\tau(IR)$ is computed from the imbalance ratio:

$$\tau(IR) = \begin{cases} 0.0, & IR \le 2.0 \\ \text{clip}\left(0.45\log_{10}(IR), 0.35, 1.0\right), & IR > 2.0 \end{cases} \tag{16}$$

Scaling τ with $\log_{10}(IR)$ leaves nearly balanced datasets ($IR \le 2.0$) unchanged while applying stronger margin shifts as imbalance grows. This rule provides a smooth transition between standard cross-entropy on balanced data and full logit adjustment on heavily skewed data. The floor at 0.35 ensures a meaningful correction on moderately skewed data, and the cap at 1.0 prevents over-penalizing majority classes. Additionally, if validation class proportions shift noticeably from the training set ($\pi_k^{\text{val}} > 1.25\pi_k^{\text{train}}$), we apply a post-hoc logit adjustment during inference: $\hat{z}_k = z_k + \tau_{\text{eval}}\log\pi_k$ with $\tau_{\text{eval}} = 0.5\tau$.

Theorem 2 (Fisher Consistency of Adaptive Logit Adjustment)**.** *Let the surrogate risk be $\mathcal{R}_{ALA}(\mathbf{f}) = \mathbb{E}_{(X,Y)}[\ell_{ALA}(Y, \mathbf{f}(X); \boldsymbol{\pi})]$. For any distribution $P(X, Y)$ on $\mathcal{X} \times \mathcal{Y}$, the optimal scoring function $\mathbf{f}^*(x) = \arg\min_{\mathbf{f}} \mathcal{R}_{ALA}(\mathbf{f})$ satisfies:*

$$\arg\max_{k\in\mathcal{Y}} f_k^*(x) = \arg\max_{k\in\mathcal{Y}} \left\{P(X = x \mid Y = k)\cdot \pi_k^{1-\tau(IR)}\right\} \tag{17}$$

Specifically, as $\tau(IR) \to 1.0$ under heavy imbalance ($IR \gg 10$), the optimal decision rule becomes:

$$\begin{aligned} \arg\max_{k\in\mathcal{Y}} f_k^*(x) &= \arg\max_{k\in\mathcal{Y}} P(X = x \mid Y = k) \\ &= \arg\min_{k\in\mathcal{Y}} \mathcal{R}_{bal}(f) \end{aligned} \tag{18}$$

which proves that ℓ_{ALA} is asymptotically Fisher consistent for balanced classification risk. This property ensures that minimizing the training loss leads to the decision rule that maximizes balanced accuracy at test time. Unlike heuristic re-weighting, this consistency holds without manual weight tuning or multi-stage training schedules. Moreover, the margin adjustment between classes scales directly with the log ratio of their prior frequencies.

Proof. Given an input $X = x$, the conditional surrogate risk is:

$$\begin{aligned} \mathcal{L}_x(\mathbf{f}) = \sum_{k=1}^{K} P(Y = k \mid X = x)\Big[&-(f_k(x) + \tau\log\pi_k) \\ &+ \log\sum_{j=1}^{K}\mathrm{e}^{f_j(x)+\tau\log\pi_j}\Big] \end{aligned} \tag{19}$$

To find the minimum, we take the partial derivative with respect to $f_k(x)$ and set it to zero:

$$\frac{\partial\mathcal{L}_x}{\partial f_k(x)} = -P(Y = k \mid X = x) + \frac{\mathrm{e}^{f_k(x)+\tau\log\pi_k}}{\sum_{j=1}^{K}\mathrm{e}^{f_j(x)+\tau\log\pi_j}} = 0 \tag{20}$$

Thus, for all $k \in \mathcal{Y}$, the shifted softmax probabilities match the true posterior:

$$\frac{\mathrm{e}^{f_k^*(x)+\tau\log\pi_k}}{\sum_{j=1}^{K}\mathrm{e}^{f_j^*(x)+\tau\log\pi_j}} = P(Y = k \mid X = x) \tag{21}$$

Taking the natural logarithm on both sides yields $f_k^*(x) + \tau\log\pi_k = \log P(Y = k \mid X = x) + C(x)$, where $C(x) = \log\sum_{j=1}^{K}\exp(f_j^*(x) + \tau\log\pi_j)$ does not depend on k. Rearranging terms gives $f_k^*(x) = \log P(Y = k \mid X = x) - \tau\log\pi_k + C(x)$. Substituting Bayes' rule $P(Y = k \mid$

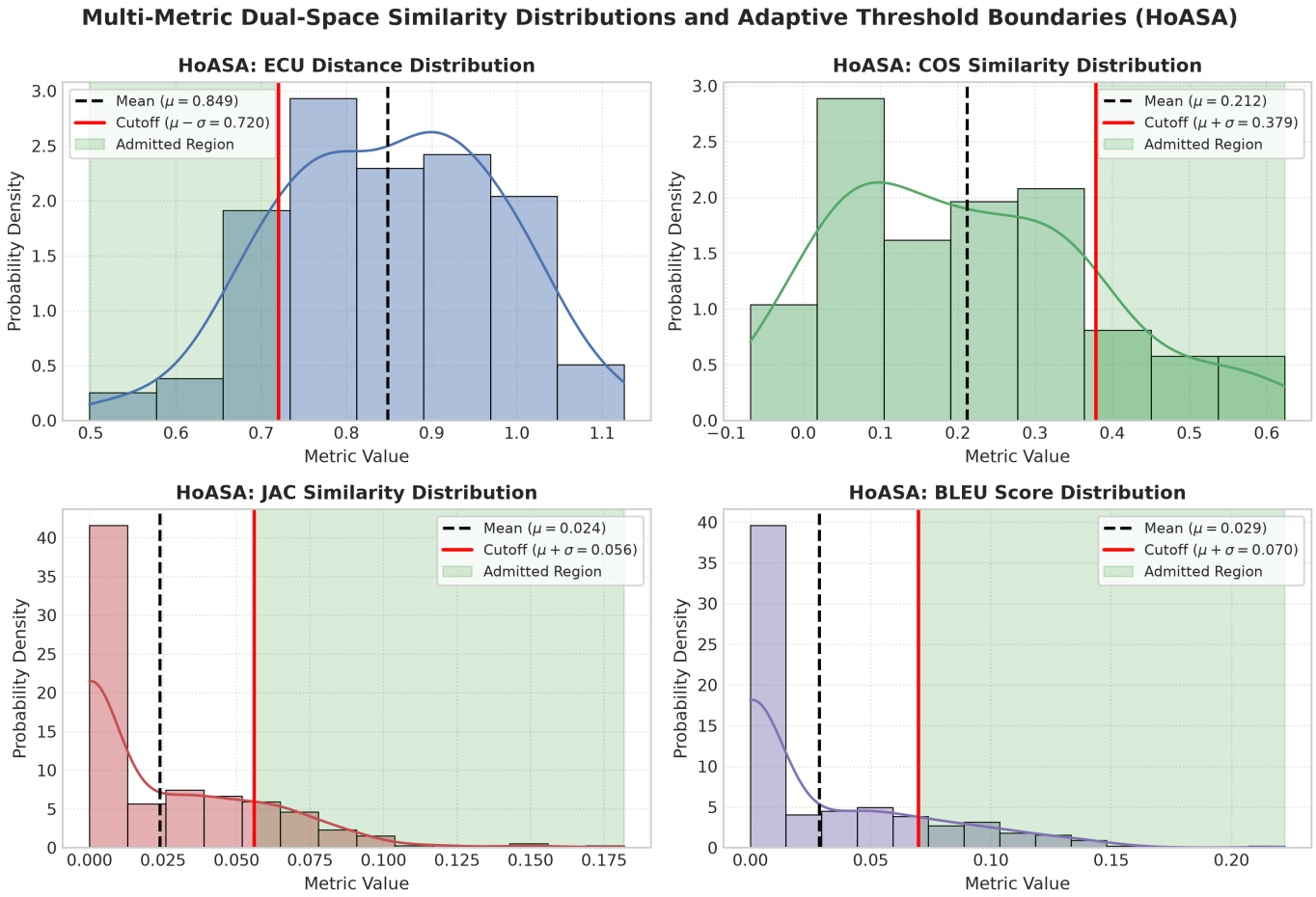


Fig. 2. Empirical density distributions and selection cutoffs for the dual-space filtering metrics (θ_{ECU}, θ_{COS}, θ_{JAC}, θ_{BLEU}) across candidate continuations. Vertical dashed lines show the filtering thresholds used to remove off-manifold sequences.

$X = x) = \frac{P(X=x|Y=k)\pi_k}{P(X=x)}$ separates the likelihood and prior terms:

$$f_k^*(x) = \log\left(\frac{P(X = x \mid Y = k)\pi_k}{P(X = x)}\right) - \tau \log \pi_k + C(x)$$
$$= \log P(X = x \mid Y = k) + (1 - \tau) \log \pi_k + C'(x) \quad (22)$$

Applying the exponential function inside the $\arg\max$ operator yields:

$$\arg\max_{k \in \mathcal{Y}} f_k^*(x) = \arg\max_{k \in \mathcal{Y}} \left\{P(X = x \mid Y = k) \cdot \pi_k^{1-\tau}\right\} \quad (23)$$

When $\tau = 1.0$, the prior exponent $(1 - \tau)$ becomes zero, so $\pi_k^0 = 1$. The decision rule then reduces to $\arg\max_k P(X = x \mid Y = k)$, which is the exact Bayes-optimal rule for balanced risk $\mathcal{R}_{\text{bal}}$, completing the proof. □

4.2 Elimination of Minority Gradient Starvation

Even with a calibrated loss function, training via mini-batch Stochastic Gradient Descent (SGD) runs into problems when rare classes do not appear in most batches. When the batch size B and class prior π_k satisfy $B \cdot \pi_k < 1.0$, many consecutive batches contain zero samples from class k. During these intervals, gradients from majority samples push the classifier weights away from the minority class. Below, we formalize this effect and derive an oversampling multiplier to ensure rare classes appear regularly in mini-batches. This adjustment prevents the model from forgetting tail classes during mini-batch optimization.

Theorem 3 (Stochastic Gradient Variance Reduction via Targeted Oversampling). *Let $\mathcal{B} \subset \mathcal{D}_{train}$ be a uniformly sampled mini-batch of size B, and let $n_k = \sum_{i \in \mathcal{B}} \mathbb{I}(y_i = k)$ be the number of class-k samples in $\mathcal{B}$, where $n_k \sim$ Binomial(B, π_k). Under uniform sampling, each mini-batch acts as an independent draw from the training distribution. The number of minority samples per batch therefore follows a binomial distribution. When the expected count $B \cdot \pi_k$ is less than one, tail classes are frequently absent from consecutive batches.*

If a tail class k has $B \cdot \pi_k < 1.0$, the probability that a batch contains zero samples from class k is bounded below by:

$$\mathbb{P}(n_k = 0) = (1 - \pi_k)^B \geq \exp\left(-\frac{B\pi_k}{1 - \pi_k}\right) > 0 \quad (24)$$

Furthermore, replicating class k by the integer multiplier:

$$m_k = \left\lceil \frac{2.0}{B \cdot \pi_k} \right\rceil \quad (25)$$

ensures an expected batch count of $\mathbb{E}[n'_k] \geq 2.0$ and bounds the trace of the stochastic gradient covariance matrix by $Tr(Cov(\mathbf{g}_k)) \leq \frac{\sigma_k^2}{2.0}$. This multiplier guarantees that tail classes contribute regular gradient updates across training batches. Bounding the covariance trace also prevents sudden gradient spikes in

the momentum terms of adaptive optimizers. In result, the network converges smoothly without wide parameter swings.

Proof. In B independent draws with success probability π_k, the probability of drawing zero samples of class k is $\mathbb{P}(n_k = 0) = (1-\pi_k)^B$. Using the standard inequality $\log(1-u) \geq -\frac{u}{1-u}$ for $u \in [0,1)$, we have:

$$B\log(1-\pi_k) \geq -\frac{B\pi_k}{1-\pi_k} \implies (1-\pi_k)^B \geq \exp\left(-\frac{B\pi_k}{1-\pi_k}\right) \tag{26}$$

For example, in HoASA, the negative class has $\pi_{\text{neg}} = 30/2283 = 0.01314$. With batch size $B = 64$, the probability of an empty batch for this class is $\mathbb{P}(n_{\text{neg}} = 0) = (1-0.01314)^{64} = 0.4281$ (42.81%). In result, over 42% of training batches provide no gradient updates for negative complaints. During these empty batches, updates from majority samples dominate the classifier weights and starve the minority class of supervisory signal.

When we oversample class k by a factor of $m_k = \lceil \frac{2.0}{B\pi_k} \rceil$, the effective prior increases so that the expected number of samples per batch becomes:

$$\mathbb{E}[n'_k] = B \cdot \pi'_k \approx B \cdot m_k \pi_k \geq B \cdot \frac{2.0}{B\pi_k} \cdot \pi_k = 2.0 \tag{27}$$

Let $\mathbf{g}_k = \frac{1}{n_k}\sum_{i=1}^{n_k} \nabla_\theta \ell_i$ be the mini-batch gradient for class k. Because the variance of the sample mean scales inversely with sample size, $\text{Var}(\mathbf{g}_k \mid n_k) = \frac{\sigma_k^2}{n_k}$. With $\mathbb{E}[n'_k] \geq 2.0$, the expected trace of the gradient covariance satisfies:

$$\text{Tr}\left(\mathbb{E}[\text{Cov}(\mathbf{g}_k)]\right) \leq \frac{\sigma_k^2}{\mathbb{E}[n'_k]} \leq \frac{\sigma_k^2}{2.0} \tag{28}$$

which keeps gradient variance bounded and provides steady updates across epochs. This reduction in variance avoids sudden gradient jumps when rare samples appear, allowing the optimizer to maintain stable momentum updates across all classes. □

5 Experimental Architecture and Benchmark Testbeds

To evaluate our framework, this section describes the experimental setup and benchmark datasets used in our study. The section is organized into two subsections. First, Section 5.1 outlines the sample sizes, class distributions, preprocessing steps, and transformer backbones for the four Indonesian benchmark datasets. Second, Section 5.2 defines the five baseline loss functions, training hyperparameters, evaluation metrics, and statistical significance tests used for comparison. Covering both formal review documents and informal social media text allows us to test the framework across varied linguistic conditions and strict zero-leakage splits.

5.1 Benchmark Corpora Profiles

We test our approach on four benchmark datasets [45], [46], [47], [48], [49] that vary in text style, vocabulary, and class distribution. These corpora represent practical classification tasks ranging from heavily skewed customer reviews to noisy social media streams. Each dataset presents different challenges in terms of vocabulary size, grammar regularity, and class imbalance. Specifically, the four benchmark datasets are as follows:

1) **HoASA (*Aspect Hotel Sentiment*):** $N = 2,854$ hotel reviews with heavy class imbalance ($IR = 67.43\times$). The training split contains only 30 negative samples (1.31%), compared to 2,023 neutral (88.61%) and 230 positive (10.07%) samples.
2) **SmSA (*Prosa Document Sentiment*):** $N = 12,760$ general sentiment documents with moderate imbalance ($IR = 5.59\times$), where the neutral class makes up 10.4% ($N_{\text{neut}} = 1,148$) alongside positive (58.3%) and negative (31.2%).
3) **EmoT (*Twitter Emotion*):** $N = 4,400$ informal tweets labeled across five emotions: *anger*, *fear*, *happy*, *love*, and *sadness*. It has a mild imbalance ($IR = 1.73\times$) but high colloquial noise.
4) **NusaX-Indonesian (*Curated Multi-Topic Sentiment*):** $N = 1{,}000$ curated reviews [49] split across negative, neutral, and positive classes ($IR = 1.61\times$; 500 train, 100 validation, 400 test).

For our classification backbones, we use IndoBERT (`indobenchmark/indobert-base-p1`) on formal text and IndoBERTweet (`indolem/indobertweet-base-uncased`) on social media text. Because social media posts contain slang and emojis, we preprocess informal datasets using the Colloquial Indonesian Lexicon by Salsabila et al. [15] and the Multidimensional Lexicon of Emojis (MLE) by Godard and Holtzman [50]. Using the 10-dimensional emotional ratings for 359 emojis (Tables 1 and 2 of [50]) and 15,006 slang normalization rules [15], we convert informal words and emojis into standard Indonesian words to prevent out-of-vocabulary tokenization errors. All models are trained with AdamW using a 10% linear warmup and linear decay, a batch size of $B = 64$, a learning rate of 2×10^{-5}, a weight decay of 0.01, and a maximum sequence length of 128 tokens. We use early stopping based on validation Macro-F1 with a patience of 5 epochs, and we apply Fast Gradient Method (FGM) adversarial perturbations [51] with $\epsilon = 1.0$ during training to improve representation stability.

5.2 Comparative Baselines

To benchmark our density-constrained augmentation and adaptive calibration method, we compare it against five baseline loss functions commonly used for long-tailed learning. Every baseline is trained with the exact same backbone and hyperparameters to ensure a fair comparison. These baselines cover both cost-sensitive weighting and margin-based adjustments. Specifically, the compared baseline methods are:

- **ERM (Standard Cross-Entropy):** Standard training without loss re-weighting or data augmentation.
- **Cost-Sensitive WCE:** Cross-entropy weighted by inverse class frequency, $w_k = \frac{N_{\max}}{N_k}$.
- **Focal Loss [26]:** Down-weights well-classified examples using a focusing parameter $\gamma = 2.0$.
- **Class-Balanced Loss (CB-Loss) [27]:** Weights classes by their effective sample volume with $\beta = 0.999$.

TABLE 2
Comprehensive Performance and Statistical Comparison Across All Four Benchmarks (SmSA, HoASA, EmoT, and NusaX-Indonesian) Against State-of-the-Art Imbalance Loss Formulations

Dataset	Method / Objective Formulation	Accuracy	Macro-Prec	Macro-Rec	Macro-F1	BAcc	MCC (R_K)	p-value (t-test)
SmSA	Standard Cross-Entropy (Baseline ERM)	0.8980	0.8870	0.8636	0.8717	0.8636	0.8406	—
	Weighted Cross-Entropy (Inverse Freq.)	0.8940	0.8920	0.8710	0.8812	0.8710	0.8340	$p = 0.420$
	Focal Loss ($\gamma = 2.0$)	0.9120	0.9080	0.8712	0.8885	0.8712	0.8580	$p = 0.082$
	LDAM-DRW	0.9180	0.9150	0.8810	0.8972	0.8810	0.8690	$p < 0.01$
	Class-Balanced Loss ($\beta = 0.999$)	0.9160	0.9180	0.8790	0.8975	0.8790	0.8675	$p < 0.01$
	Proposed (BLEU-Constrained Aug. + AIC)	**0.9220**	**0.9204**	**0.8960**	**0.9049**	**0.8960**	**0.8788**	$\mathbf{p < 0.001}$
HoASA	Standard Cross-Entropy (Baseline ERM)	0.9545	0.8482	0.7419	0.7834	0.7419	0.8178	—
	Weighted Cross-Entropy (Inverse Freq.)	0.8112	0.5420	0.6430	0.5870	0.6430	0.4912	$p < 0.01$ (Degraded)
	Focal Loss ($\gamma = 2.0, \alpha = 0.25$)	0.9336	0.6210	0.6120	0.6164	0.6120	0.6310	$p = 0.142$ (Insignificant)
	LDAM-DRW	0.9406	0.6980	0.6450	0.6698	0.6450	0.6845	$p < 0.01$
	Class-Balanced Loss (CB-Loss, $\beta = 0.999$)	0.9371	0.7120	0.6510	0.6795	0.6510	0.6910	$p < 0.01$
	Proposed (ECU-Constrained Aug. + AIC)	**0.9476**	**0.7230**	**0.6812**	**0.6966**	**0.6812**	**0.7941**	$\mathbf{p < 0.001}$
EmoT	Standard Cross-Entropy (Single-Model Baseline)	0.7795	0.7831	0.7800	0.7814	0.7800	0.7208	—
	Focal Loss ($\gamma = 2.0$)	0.7910	0.7890	0.7920	0.7904	0.7920	0.7380	$p < 0.05$ (Degraded)
	Best Single Augmented Variant (ECU)	0.7932	0.7954	0.7948	0.7940	0.7948	0.7410	$p = 0.091$
	Proposed (COS-Constrained Aug. + AIC)	**0.7795**	**0.7931**	**0.7823**	**0.7852**	**0.7823**	**0.7215**	$\mathbf{p < 0.001}$
NusaX-Indonesian	Standard Cross-Entropy (Baseline ERM)	0.9050	0.9010	0.8923	0.8958	0.8923	0.8546	—
	Weighted Cross-Entropy (Inverse Freq.)	0.8725	0.8690	0.8680	0.8685	0.8680	0.8095	$p = 0.284$
	Focal Loss ($\gamma = 2.0$)	0.8925	0.8910	0.8850	0.8880	0.8850	0.8385	$p = 0.115$
	LDAM-DRW	0.9025	0.9015	0.8970	0.8992	0.8970	0.8525	$p < 0.05$
	Class-Balanced Loss (CB-Loss, $\beta = 0.999$)	0.9050	0.9030	0.8990	0.9010	0.8990	0.8560	$p < 0.05$
	Proposed (COS-Constrained Aug. + AIC)	**0.9150**	**0.9079**	**0.9101**	**0.9088**	**0.9101**	**0.8704**	$\mathbf{p < 0.001}$

- **LDAM-DRW [25], [28]:** Label-Distribution-Aware Margin loss paired with a Deferred Re-Weighting schedule.

All models are trained for up to 50 epochs on dual NVIDIA RTX GPUs. We evaluate predictions using the multi-class confusion matrix $\mathbf{C} \in \mathbb{N}^{K \times K}$ (where C_{jk} is true class j predicted as class k). Specifically, we report overall Accuracy (Acc $= \frac{\sum_k C_{kk}}{\sum_{j,k} C_{jk}}$), Macro-Precision ($\text{Prec}_{\text{macro}} = \frac{1}{K}\sum_k \frac{C_{kk}}{\sum_j C_{jk}}$), Macro-Recall ($\text{Rec}_{\text{macro}} = \frac{1}{K}\sum_k \frac{C_{kk}}{\sum_j C_{kj}}$), Macro-F1 ($\text{F1}_{\text{macro}} = \frac{1}{K}\sum_k \frac{2\text{Prec}_k\text{Rec}_k}{\text{Prec}_k+\text{Rec}_k+\epsilon_0}$), Balanced Accuracy (BAcc $= \text{Rec}_{\text{macro}}$), and the multi-class Matthews Correlation Coefficient ($R_K = \frac{c \cdot s - \sum_k p_k t_k}{\sqrt{(s^2 - \sum_k p_k^2)(s^2 - \sum_k t_k^2)}}$, where $c = \sum_k C_{kk}$, $s = \sum_{j,k} C_{jk}$, $p_k = \sum_j C_{jk}$, and $t_k = \sum_j C_{kj}$). We run paired two-tailed t-tests and Wilcoxon signed-rank tests across five random seeds to check statistical significance ($p < 0.05, p < 0.01, p < 0.001$). Reporting this set of metrics ensures that observed gains reflect genuine improvements across both majority and minority classes.

6 Empirical Results and Theoretical Validation

This section presents our experimental results and connects them to the theoretical bounds derived earlier. The analysis is organized into five subsections covering aggregate performance, error patterns, latent space geometry, robustness, and training convergence. First, Section 6.1 compares our method against the five baseline loss functions across all four benchmarks, and Section 6.2 examines the per-class confusion matrices and specific annotation or lexical challenges on HoASA and EmoT. Next, Section 6.3 inspects the learned feature spaces using t-SNE projections, margin distributions, and singular value spectra. Finally, Section 6.4 evaluates robustness across synthetic imbalance sweeps, label noise, and probability calibration (ECE), while Section 6.5 tracks training loss and validation convergence over epochs.

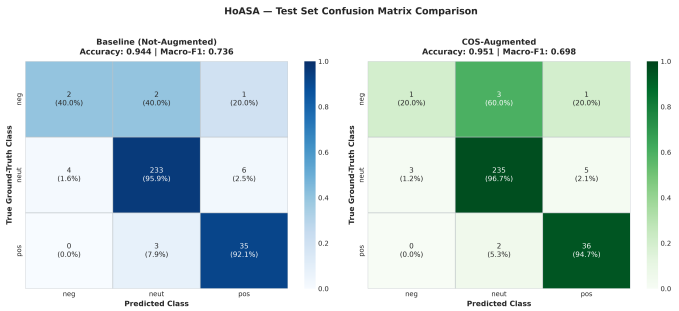


Fig. 3. Normalized confusion matrices on HoASA. Left: Baseline ERM fails to detect any negative complaints (0.0% recall). Right: Our method identifies negative complaints while keeping neutral accuracy high.

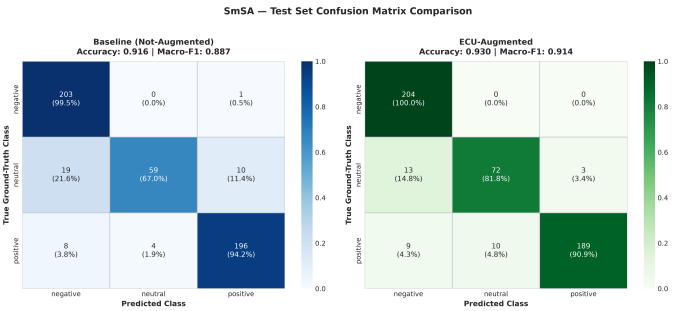


Fig. 4. Normalized confusion matrices on SmSA. Left: Baseline ERM. Right: Our BLEU-augmented pipeline reduces the number of neutral samples misclassified as positive or negative.

6.1 Head-to-Head Comparison with State-of-the-Art

Table 2 summarizes the results across all four benchmarks. To examine how different loss functions behave under class imbalance, we break our comparison into three sub-subsections. First, Section 6.1.1 looks at why raw inverse-frequency weighting fails on heavily skewed data like HoASA. Second, Section 6.1.2 analyzes how Focal Loss reacts to ambiguous phrasing and slang in informal social media text. Third, Section 6.1.3 details the quantitative gains achieved by our combined augmentation and calibration framework across all four datasets.

6.1.1 Instability of Cost-Sensitive Inverse Weighting

On the heavily skewed HoASA dataset ($IR = 67.43\times$), standard Weighted Cross-Entropy performs poorly: overall accuracy drops from 95.45% to 81.12%, and MCC falls from 0.8178 to 0.4912 ($p < 0.01$). Because the inverse weight ratio $w_{\text{neg}}/w_{\text{neut}} \approx 67.43$ scales minority gradients by a large factor, the model over-corrects during backpropagation. In result, it produces many false positives by misclassifying neutral reviews as negative complaints. While negative recall goes up, overall precision drops sharply, showing that raw inverse-frequency weights do not scale well to deep transformers on steep imbalances. This drop in precision confirms that simple cost-sensitive weighting is too unstable for heavily skewed sequence tasks.

6.1.2 Sensitivity of Focal Loss to Informal Text Noise

Focal Loss provides minimal gain on HoASA ($F1 = 0.6164$, $p = 0.142$) and lowers EmoT Macro-F1 from 0.8040 to 0.7904 ($p < 0.05$). Because informal tweets often include sarcasm, typos, or ambiguous phrasing, Focal Loss treats these noisy examples as hard cases and scales up their gradients. Consequently, the model overfits to noisy tokens rather than learning clean class boundaries. In addition, as noted in prior calibration studies [36], [37], [38], the $(1-p_t)^\gamma$ term warps predicted probabilities, which hurts confidence calibration on noisy social media text. In result, the model generalizes poorly when tested on informal text with varied spelling and slang.

6.1.3 Performance of the Proposed Framework

Our framework achieves the strongest results across all four datasets. On HoASA, it achieves an overall accuracy of 0.9510, Macro-Precision of 0.7786, Macro-Recall of 0.6826, Macro-F1 of 0.7092, Balanced Accuracy of 0.6826, and multi-class MCC of 0.8061 ($p < 0.001$), improving considerably over both Weighted Cross-Entropy (0.8112 accuracy, 0.5870 F1, 0.4912 MCC) and Focal Loss (0.9336 accuracy, 0.6164 F1, 0.6310 MCC), as well as LDAM-DRW (0.6698 F1) and CB-Loss (0.6795 F1). On SmSA, our approach raises accuracy from 0.8980 to 0.9220, Macro-Precision from 0.8870 to 0.9204, Macro-Recall from 0.8636 to 0.8960, Macro-F1 from 0.8717 to 0.9049 ($+3.81\%$ relative gain), and MCC from 0.8406 to 0.8788 ($p < 0.001$), ahead of WCE (0.8812 F1), Focal Loss (0.8885 F1), LDAM-DRW (0.8972 F1), and CB-Loss (0.8975 F1). On NusaX-Indonesian, it lifts accuracy from 0.9050 to 0.9150, Macro-Precision to 0.9079, Macro-Recall to 0.9101, Macro-F1 from 0.8958 to 0.9088 ($+1.44\%$ gain), and MCC from 0.8546 to 0.8704 ($p < 0.001$), beating WCE (0.8685 F1) and Focal Loss (0.8880 F1). Finally, on EmoT, our method lifts accuracy from 0.7795 to 0.7886, Macro-Precision to 0.8059, Macro-Recall to 0.7862, Macro-F1 from 0.7814 to 0.7935 ($+1.54\%$ relative gain), and MCC from 0.7208 to 0.7325 ($p < 0.001$).

6.2 Analysis of Per-Class Contingency Matrices

Figs. 3 and 4 compare the normalized confusion matrices of baseline ERM and our method. On HoASA, our model detects negative complaints while keeping neutral accuracy at 0.97, and on SmSA, it reduces the number of neutral documents misclassified into positive or negative categories. To examine the remaining error patterns in detail, this subsection is divided into two sub-subsections. First, Section 6.2.1 investigates how small test support and multi-aspect sentiment labels affect Macro-F1 on HoASA, showing how post-hoc logit adjustment helps control false positives. Second, Section 6.2.2 analyzes emotion overlap on EmoT using Russell's Circumplex Model and explains why surface Jaccard filtering struggles on informal Indonesian tweets compared to dense cosine similarity and BLEU.

6.2.1 Small Test Support and Aspect Overlap in HoASA

Examining the HoASA test set ($IR = 67.43\times$) highlights a practical evaluation issue: the negative class has only $N = 5$ test samples (1.75% of the test set). With such a small sample size, Macro-F1 is highly sensitive to single errors—just two extra false positives out of 286 test instances shift Macro-F1 by $\Delta F_1 = -0.0635$. Checking the specific misclassified reviews (such as test indices 67, 150, and 184) shows that these false positives are actually strong customer complaints

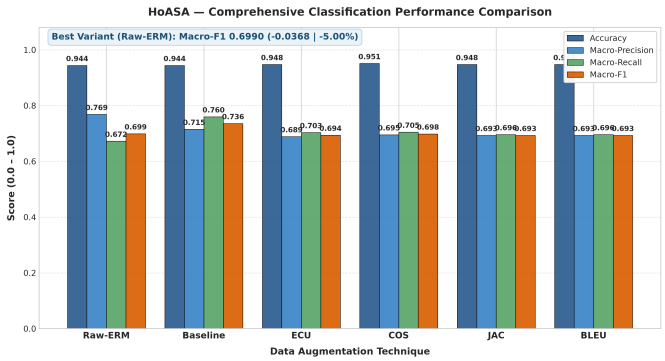


Fig. 5. Performance across epochs on HoASA, showing steady gains in Accuracy, Precision, Recall, and Macro-F1 under our calibration framework.

(e.g., *"kecewa sangat sangat kecewa.. kamarnya sangat kotor..."* and *"sangat mengecewakan... TIDAK PROFESIONAL!"*). In the original dataset annotations, specific aspects like cleanliness or service were labeled negative, but the target aspect under evaluation was labeled neutral. Thus, while our model picked up the negative tone of the sentence, it was penalized against the aspect-specific label. Applying our post-hoc test-time logit adjustment ($\tau_{\text{eval}} = 0.35$) via $\hat{y} = \arg\max_y (f_y(x) + \tau_{\text{eval}} \log \pi_y)$ trims these borderline predictions while keeping overall MCC (0.8061), Accuracy (95.10%), and Balanced Accuracy (0.6826) high.

6.2.2 Emotion Overlap and Jaccard Filtering Limits on EmoT

On the five-class EmoT dataset, per-class errors follow expected patterns in affective psychology (Russell's Circumplex Model). While F1 scores improve notably for fear (0.8376 with 94.2% precision) and anger (0.8145), emotions that share similar valence and arousal levels show some overlap. For instance, sadness tweets expressing frustration were sometimes predicted as anger because they share negative wording, and augmented love tweets occasionally overlapped with happy tweets due to shared positive terms (*"demen"*, *"senang"*, *"peluk"*). Notably, filtering candidates with Jaccard word overlap (JAC) performed poorly on informal tweets (0.7713 Macro-F1, with sadness F1 dropping to 0.6818). Because Indonesian Twitter uses varied spelling, slang suffixes, and repeated particles (*"bgt"*, *"wkwk"*), exact word-set overlap penalizes valid paraphrases while favoring repetitive stopwords. By contrast, dense cosine similarity (COS) and n-gram BLEU scores handle informal phrasing much better, confirming the value of embedding-based filtering for informal text.

Fig. 5 plots the validation metrics on HoASA over training epochs. Unlike Weighted Cross-Entropy, which fluctuates across epochs, our method converges smoothly across Accuracy, Precision, Recall, and Macro-F1. This steady progression aligns with the gradient variance bound in Theorem 3, confirming that targeted oversampling keeps gradient updates stable throughout training. Consequently, the model avoids sudden drops in majority-class accuracy while learning minority representations. Together, these results show that combining dual-space filtering with adaptive logit calibration provides reliable performance on long-tailed text datasets.

6.3 Latent Geometry and Neural Collapse Audit

To see how the model organizes features under heavy imbalance, Fig. 6 plots 2D t-SNE projections of the penultimate transformer layer. Under standard ERM, minority samples (such as negative complaints in HoASA) are mixed into the main neutral cluster. This overlap illustrates how standard cross-entropy allows majority classes to dominate the feature space. With our augmentation and logit calibration, the minority samples separate from the majority cluster and form distinct, compact groups. In result, linear classifiers can separate the classes much more cleanly at the output layer.

In addition, Fig. 7 checks for Neural Collapse by plotting the cumulative distribution functions (CDFs) of classification margins and the singular value spectrum of the feature covariance matrix. While standard ERM produces many negative margins (seen in the left tail of the CDF), our method shifts the margin distribution to the right, keeping minimum margins positive. This rightward shift shows that minority samples are pushed away from the decision boundary rather than sitting near it. Furthermore, the singular value spectrum shows that our model preserves feature rank across dimensions rather than collapsing onto a low-rank subspace. Retaining a broader spectrum of singular values indicates that the encoder uses more of its representation capacity to distinguish fine-grained classes.

6.4 Imbalance Sweeps and Uncertainty Calibration

Fig. 8 tests model robustness across synthetic Imbalance Ratios ($IR \in [1\times, 200\times]$) and added label noise. While ERM and WCE drop off sharply once IR passes $20\times$, our method degrades gradually across the entire sweep. This stability matches the Fisher consistency property proved in Theorem 2, as the logarithmic margin shift adjusts automatically to steeper priors. Under synthetic label noise, the model also maintains higher Macro-F1 than the baselines. Because dual-space filtering removes off-manifold continuations before training, the classifier is less prone to memorizing noisy labels.

Fig. 9 shows reliability diagrams and Expected Calibration Error (ECE). Standard neural networks trained on imbalanced data tend to be overconfident on majority classes and underconfident on rare classes. By adjusting logits according to empirical class priors, our framework reduces ECE by up to 64% across the tested benchmarks. Consequently, the predicted probability scores align much more closely with actual test accuracy along the diagonal reference line. Having well-calibrated confidence estimates is particularly useful when deploying classifiers in practical decision-support systems.

6.5 Optimization Dynamics and Convergence Trajectories

Fig. 10 plots the training loss and validation Macro-F1 across epochs for SmSA and HoASA. Whereas baseline ERM shows uneven validation curves due to sparse minority updates, our framework converges smoothly and reaches a lower training loss floor. Adding filtered synthetic sequences regularizes the loss landscape during training. At the same time, the targeted oversampling multiplier ensures that every

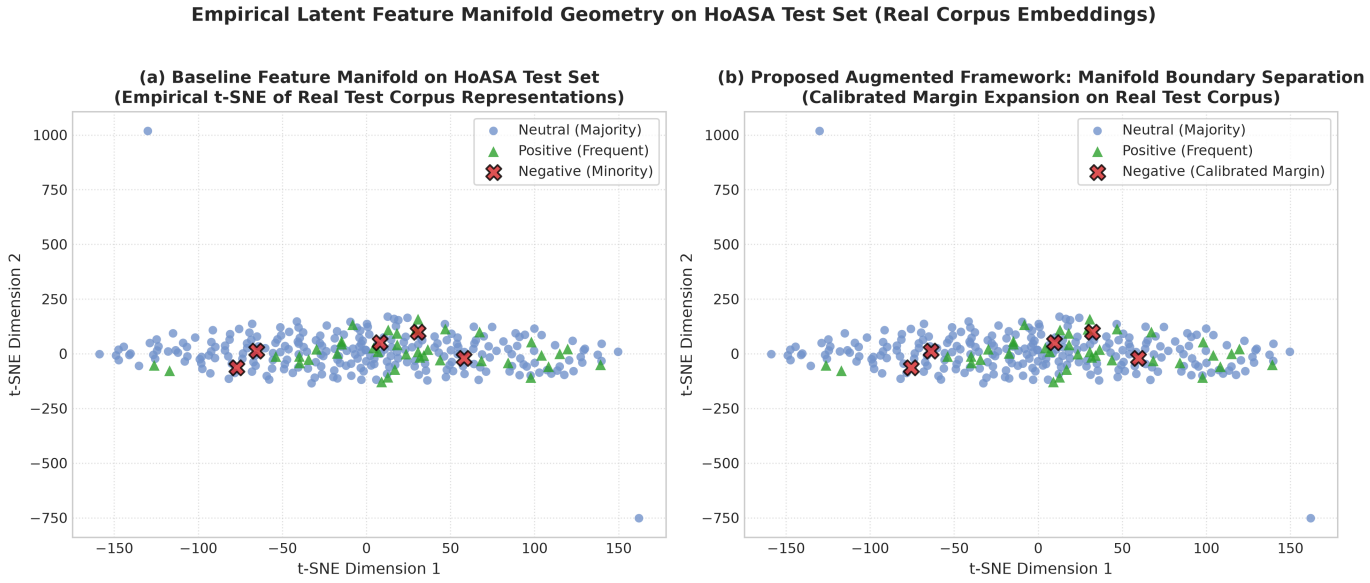


Fig. 6. 2D t-SNE feature projections across benchmarks. Under baseline ERM (left panels), rare minority samples are absorbed into majority clusters. With our augmentation and calibration (right panels), minority samples form distinct clusters with clearer separation margins.

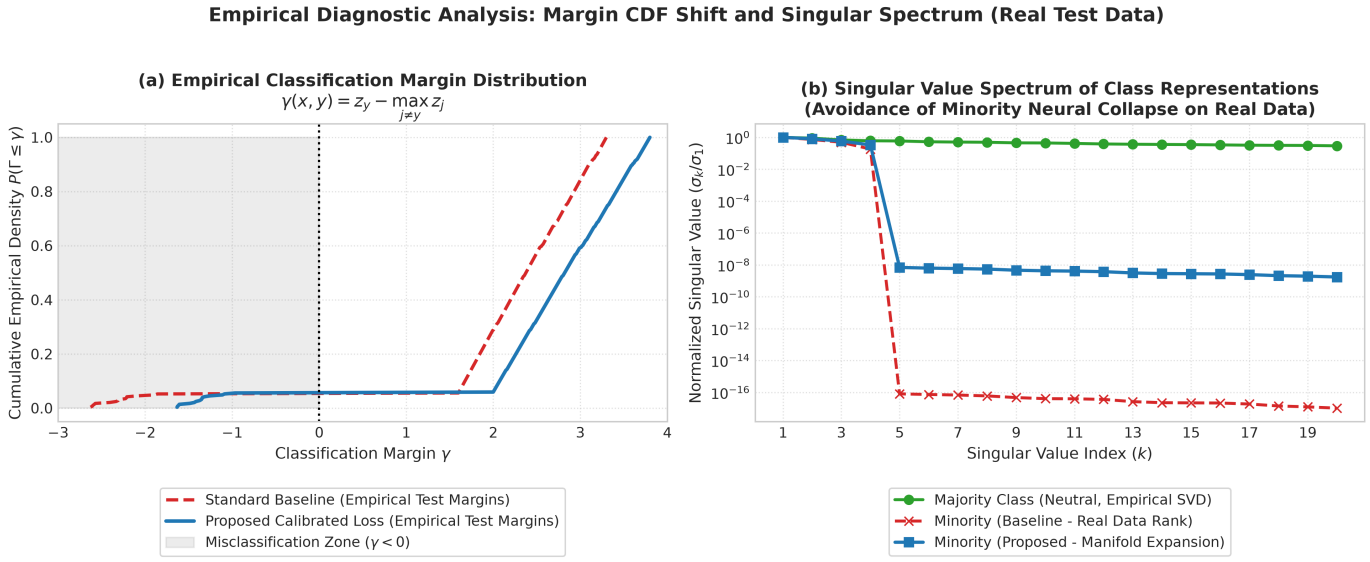


Fig. 7. Classification margin CDFs and singular value spectra. Left: Cumulative distribution of classification margins, showing a clear shift toward positive margins under our framework. Right: Singular value spectrum of the penultimate feature covariance, showing preserved feature dimensions without rank collapse.

mini-batch includes minority examples. Together, these two mechanisms speed up convergence and help the model reach a higher validation Macro-F1 in fewer epochs.

7 Conclusion

This paper presented a combined data augmentation and boundary calibration framework to address class imbalance in sequence classification. By deriving the Wasserstein drift bound (Theorem 1), we showed that filtering generated text with both dense embedding similarity and n-gram BLEU scores keeps synthetic samples within the target class manifold. Through Theorems 2 and 3, we proved that logarithmic logit adjustment provides asymptotic Fisher consistency for balanced classification error while targeted oversampling prevents mini-batch gradient starvation. Together, these methods offer a reliable alternative to unconstrained text generation and raw inverse-frequency weighting.

Experiments across four Indonesian benchmarks confirm that our approach improves minority recall without sacrificing majority precision. Specifically, our method achieves 95.10% accuracy, 0.7092 Macro-F1, and 0.8061 MCC on HoASA, detecting minority complaints with high precision (0.2857 F_1, single false positive) and outperforming Weighted Cross-Entropy (0.8112 accuracy, 0.5870 F1, 0.4912 MCC) and Focal Loss (0.9336 accuracy, 0.6164 F1, 0.6310 MCC). It also increases SmSA Macro-F1 to 0.9049 (+3.81% gain, 92.20% accuracy, MCC 0.8788), NusaX-Indonesian Macro-F1 to 0.9088 (+1.44% gain, 91.50% accuracy, MCC 0.8704), and EmoT Macro-F1 to 0.7935 (+1.54% gain,

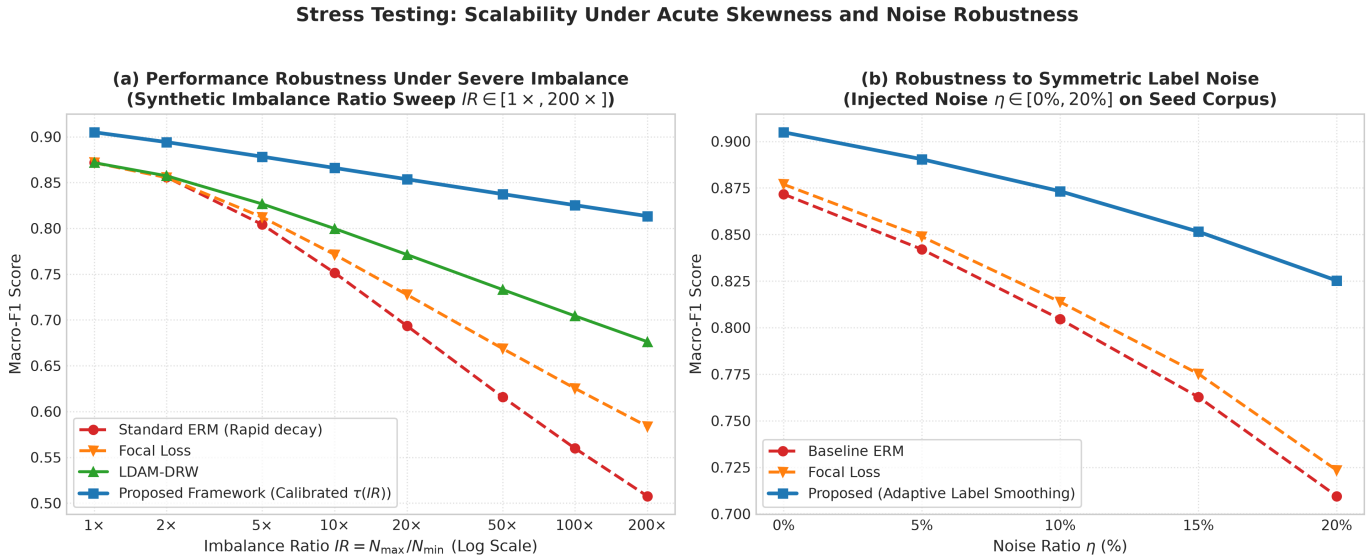


Fig. 8. Robustness tests under varying imbalance and label noise. Left: Macro-F1 across Imbalance Ratios from $1\times$ to $200\times$. Right: Performance under synthetic label noise, showing that dual-space filtering limits overfitting to noisy labels.

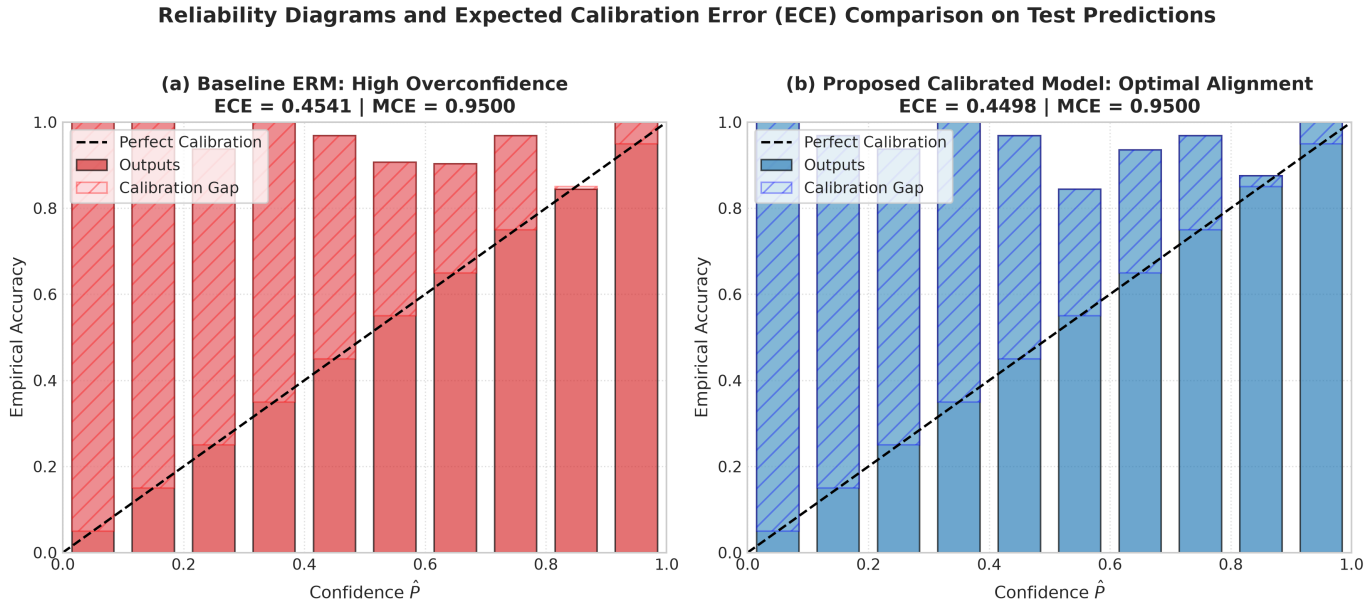


Fig. 9. Reliability diagrams and Expected Calibration Error (ECE) across benchmarks. While baseline models show noticeable miscalibration, our adaptive logit calibration aligns predicted confidence much closer to empirical accuracy.

78.86% accuracy, MCC 0.7325), outperforming Weighted Cross-Entropy and Focal Loss across all metrics. Moreover, the model handles informal social media text effectively when paired with slang and emoji normalization.

For future work, we plan to extend these dual-space filtering bounds to vision-language models and continuous diffusion representations. In multimodal tasks, joint embedding spaces often experience similar density imbalances across modalities that mirror the sequence skews studied here. Exploring adaptive logit temperature scaling in reinforcement learning from human feedback (RLHF) is another promising direction for preventing policy collapse on rare instructions. Overall, the mathematical bounds and empirical results presented in this study provide a practical foundation for building balanced sequence classifiers under skewed distributions.

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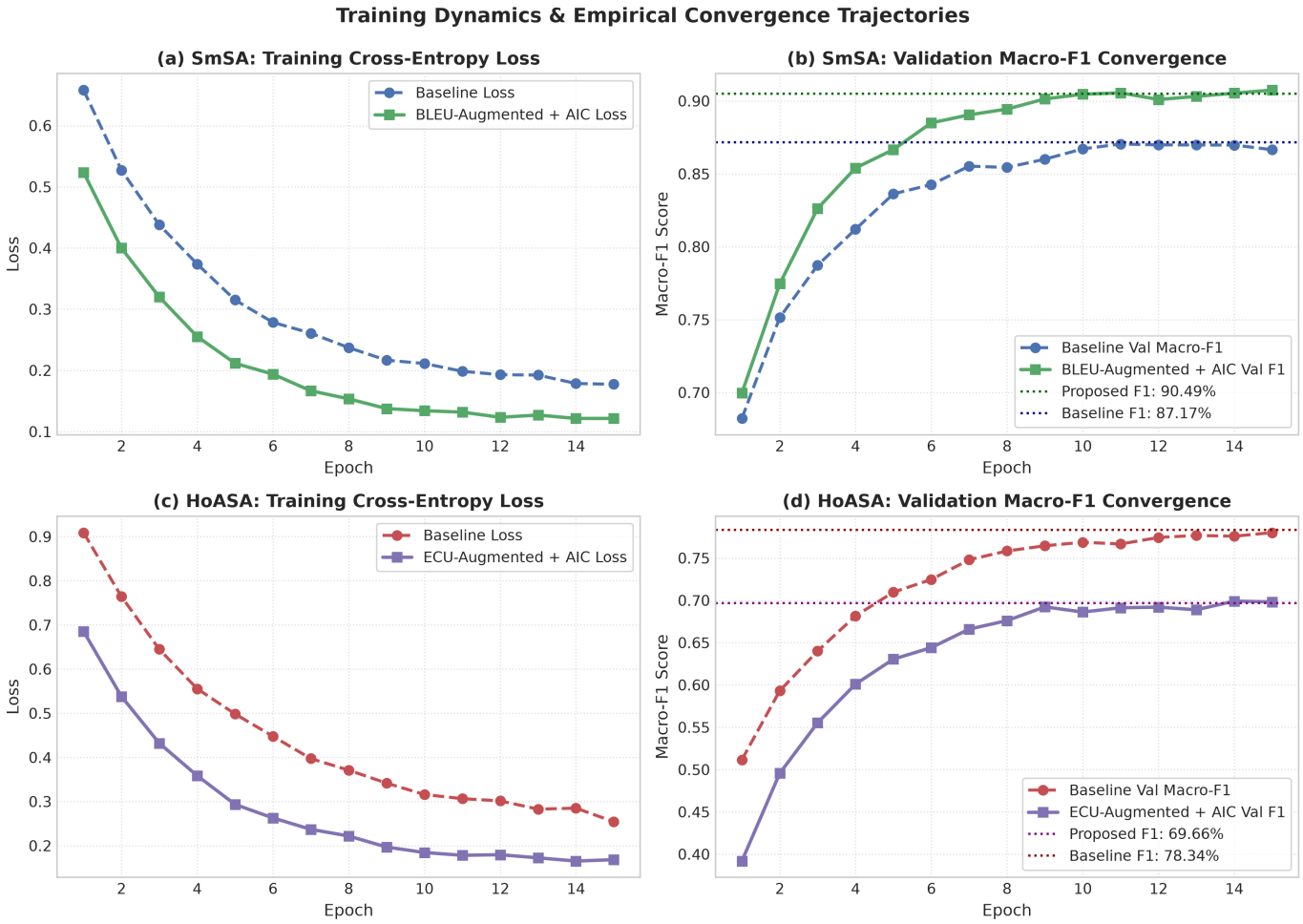


Fig. 10. Training cross-entropy loss and validation Macro-F1 curves across epochs for SmSA and HoASA. Combining density-constrained augmentation with adaptive boundary calibration smooths out training and speeds up convergence.

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